PLANETFALL

THE FLOYD RESCUE

A maximum-score route to the Cryo-Anteroom ending

SCORE	COMMANDS	DEATHS	INSTRUCTIONS
80 / 80	**546**	**0**	**488,709**

Verified story: Z-machine V3, Release 37, Serial 851003. The route reaches the successful Cryo-Anteroom ending with the maximum 80 points and no player deaths. It was replayed from a fresh game with deterministic seed 1.

How to use this guide

Follow the numbered stages in order. Commands are shown in Courier. The instructions condense the full regression route, including timed waits and incidental checks. Repeat commands only where stated. Keep the canteen and follow the rest and food directions; the late-game relay sequence leaves little room for extra turns.

Route checkpoints

The replay consumes 546 command inputs and 488,709 VM instructions. The final score is 80/80 at the successful ending, where Veldina confirms the planetary systems are repaired and Floyd returns. The opcode inventory records each distinct executed instruction with its count, first sequence, and first byte address.

Story fingerprint: SHA-256 FB6023561FD4A15ABA5D921FC9213BACC2D99C513927A96AAA96E03EA86D9CA6

Route fingerprint: SHA-256 437621B2651098DB00DFE96723EF348B2DC0785F2114BD548CA8C9B85B88A7B1

Opcode sequence numbers are 1-based; program counters are hexadecimal byte addresses.

01. Escape the Feinstein

Wait for the emergency, enter the escape pod webbing, and let the pod reach Resida. Leave the pod and take the survival kit.

```
wait
g (repeat until the pod door opens)
west
enter webbing
wait
g (repeat through landing)
exit webbing
take survival kit
open door; out; up
```

At the Crag, climb the stairway to the settlement. Save the kit's food for the moments when hunger or thirst warns you.

02. Gather tools and access cards

Collect the good bedistor in Storage East, plus the laser, flask, pliers, and metal bar from the Tool Room. Use the bar at the crevice for the key. Use that key to enter the ladder area, bridge the rift, and collect the upper-elevator and shuttle cards from the desks. Search and activate the intact robot in the Robot Shop to obtain the lower-elevator card; Floyd will come to life later.

```
go up to the courtyard
north, northeast, east three times (Corridor Junction)
drop survival kit
south, east (Storage East); take good bedistor
west, south twice, southwest (Tool Room)
drop brush and brochure; take laser, flask, pliers, and bar
northeast, north four times; hold bar over crevice
south to Corridor Junction; drop all but key
west twice; unlock padlock; take and drop padlock and key
open door; north; take ladder
south, east twice, north twice; extend ladder over rift
north, west; open desk; get all from desk
west; open wooden desk; take shuttle card
east twice, south six times, southeast; search robot; activate robot
```

03. Repair communications

Eat the red goo from the survival kit, then use the upper elevator to inspect the Comm Room enunciator. Note the flashing color. At the Machine Shop, fill the flask with that color and pour it into the Comm Room funnel. Repeat until all lights go dark and the transmission begins. This route used green, rested, then yellow.

```
northwest, north three times; take survival kit
open kit; eat red goo; drop kit; take flask
east to Elevator Lobby; press blue button; wait for the door; north
slide upper card through slot; press up; wait for doors
south, northeast (Comm Room); note the light color
southwest, north; press down; wait; south
west, south four times (Machine Shop)
put flask under spout; press matching color; take flask
north four times, east, north; use upper card; press up
wait for doors; south, northeast; empty flask into funnel
repeat the chemical fetch-and-pour cycle until transmission begins
```

If the Comm Room requires a third chemical, retrieve the survival kit at the Corridor Junction, sleep in Dorm D, eat the green goo, and continue.

04. Reach Lawanda

Take the lower elevator to the shuttle level. Wait until the shuttle reaches speed 60, pull the lever twice, then wait for it to dock. Follow the platform route to the lab pocket for the teleportation card and fresh battery.

```
southwest, north; use upper card; press down; wait for doors
south; drop upper card and flask
press red button; wait for lower elevator
south; slide lower card through slot; press down; wait for doors
north, east, south, east; slide shuttle card through slot; push lever
wait for speed 60; pull lever twice; wait until docked
west, north; east twice, southeast, east three times, south
open lab pocket; take teleportation card and fresh battery
```

05. Kitchen, rest, and system repairs

After reaching Lawanda, teleport to Booth 2 and retrieve the laser, pliers, and good bedistor from the earlier rooms; install the fresh battery. Return to Course Control and replace the fused bedistor. Wait in the Repair Room for Floyd to arrive, then have him retrieve the shiny board. Replace the second board in Planetary Defense. Treat the disease, fill a canteen, and return to Booth 3.

```
from Booth 3, use the teleportation card; press 2
west twice, south three times, southwest (Tool Room); take laser and pliers
retrieve the good bedistor from Storage East; install fresh battery
return to Booth 2; teleport to Booth 3; west, north twice (Course Control)
open lid; remove fused bedistor with pliers; put good bedistor in cube
drop pliers and fused bedistor
south, west twice, north (Repair Room); wait for Floyd
floyd, north; after he returns, floyd, get board
south, east, north (Planetary Defense); open panel
take second board; put shiny board in panel; drop fried board
south, west, northwest (Infirmary); take medicine bottle; open and drink it
return to Booth 3; teleport to Booth 2
west twice to Corridor Junction; south three times, southwest (Tool Room)
take kitchen access card; northeast, north three times, west twice, south (Mess Hall)
take canteen; use kitchen card; enter Kitchen; fill canteen with brown liquid
north; drop kitchen card; north, east four times to Booth 2
teleport to Booth 3; west, south twice to Computer Room
```

Use room descriptions to confirm junctions. Keep a full canteen for the final stages.

06. Get the miniaturization card

Wait in the Computer Room until Floyd arrives and notices the broken computer. Go northeast to the Main Lab, open the Bio-Lock, and enter its east chamber. Look through the eastern window. Let Floyd volunteer, open and close the Bio-Lab door, wait for three quick knocks, then open and close it again. Take the card he leaves behind.

```
wait for Floyd to recognize the broken computer
northeast; open bio-lock door; southeast; east
look through window; let Floyd volunteer
open door; close door; wait for three quick knocks
open door; close door; take miniaturization card
```

Before entering Station 384, refill the canteen and sleep in Dorm D. After waking, drink if hungry or thirsty; then return to the Tool Room for the laser and both cards. Go through Booth 3 to the Miniaturization Booth.

```
west; open bio-lock door; west twice; south twice
return to Booth 2, then go southwest to Tool Room
drop teleporter card, mini card, and laser; take kitchen access card
reach Mess Hall; take canteen; use kitchen card; fill canteen
return to Mess Hall; drop kitchen card; north; east; north (Dorm D)
lie down; wait; get up; take canteen; open and drink; close
return to Tool Room; take laser, teleportation card, mini card
return to Booth 2; teleport to Booth 3; west, south three times
```

07. Repair the tiny relay

Activate the miniaturization booth and enter Station 384. Destroy the relay speck with the laser set to 1. Set the laser to 6 for the microbe, fire until the laser feels quite hot, then throw it. Leave immediately.

```
slide miniaturization card through slot; type 384
east, north twice; look into relay
set dial to 1; shoot speck with laser; g until vaporized
set dial to 6; south; shoot microbe with laser
g until the laser feels quite hot; throw laser at microbe
south, west
```

08. Reach the Cryo-Anteroom

Drink from the canteen before the final escape. Wear the gas mask before pressing the red button; then cross the Bio-Lab and take the cryo-elevator to the anteroom.

```
open canteen; drink brown liquid; close canteen
north; open desk; take gas mask; wear gas mask
press red button; open door; west
open lab door; west twice
open bio-lock door; west three times; south twice
press elevator button; wait until doors open; north
```

The successful ending confirms the cure and repaired systems, and Floyd returns with the sequel items. The run ends with 80/80 points and zero deaths.

Opcode inventory

All distinct opcodes observed during the complete fresh replay are listed here. Counts and first execution locations come from the 488,709-instruction JSONL trace.

Form	Opcode	Mnemonic	Count	First seq.	First PC
Long	0x01	je	70,735	6	0x005C63
Long	0x02	jl	4,218	394	0x00ACBE
Long	0x03	jg	7,778	356	0x00A7B1
Long	0x04	dec_chk	3,439	328	0x00A4CB
Long	0x05	inc_chk	13,427	172	0x00C213
Long	0x06	jin	3,878	199	0x00C225
Long	0x07	test	5,576	351	0x00A7AD
Long	0x09	and	2,125	357	0x00A7B5
Long	0x0A	test_attr	13,187	164	0x0056EC
Long	0x0B	set_attr	184	213	0x00BB53
Long	0x0C	clear_attr	84	4,404	0x0078B8
Long	0x0D	store	33,939	155	0x0056CD
Long	0x0E	insert_obj	596	2,258	0x006EBF
Long	0x0F	loadw	47,727	20	0x005C7D
Long	0x10	loadb	11,217	173	0x00C21A
Long	0x11	get_prop	4,189	223	0x00BB7D
Long	0x12	get_prop_addr	5,900	446	0x00D401
Long	0x14	add	32,473	4	0x005C5B
Long	0x15	sub	5,957	7	0x005C67
Long	0x16	mul	1,021	367	0x00A61B
Long	0x17	div	2,521	4,795	0x00B985
Short	0x00	jz	74,131	8	0x005C6B
Short	0x00	rtrue	2,501	234	0x0063F4
Short	0x01	get_sibling	4,157	258	0x00BD23
Short	0x01	rfalse	2,680	4,857	0x00B8DF
Short	0x02	get_child	4,356	240	0x00BBBA
Short	0x02	print	1,414	166	0x00C167
Short	0x03	get_parent	5,546	217	0x00BB5F
Short	0x03	print_ret	369	235	0x006668
Short	0x04	get_prop_len	3,249	448	0x00D408
Short	0x05	inc	1,603	2,689	0x006E92
Short	0x06	dec	36	791	0x0118EB
Short	0x07	print_addr	1	30,985	0x00B070
Short	0x08	ret_popped	3,032	273	0x00BBD6
Short	0x09	remove_obj	20	3,960	0x007A86
Short	0x0A	print_obj	446	216	0x00BB5D
Short	0x0B	new_line	1,485	198	0x00C224
Short	0x0B	ret	5,200	11	0x005C7B
Short	0x0C	jump	46,037	23	0x005C8B
Short	0x0D	print_paddr	245	2,698	0x006EA4
Short	0x0E	load	2	449,901	0x00C4B6
Variable	0x00	call	27,175	2	0x00567A

Form	Opcode	Mnemonic	Count	First seq.	First PC
Variable	0x01	je	16,918	253	0x00BCCB
Variable	0x01	storew	12,406	10	0x005C76
Variable	0x02	jl	3	127,967	0x008282
Variable	0x02	storeb	538	1	0x005675
Variable	0x03	jg	36	127,971	0x008295
Variable	0x03	put_prop	22	4,323	0x00BE20
Variable	0x04	sread	546	335	0x00A4DF
Variable	0x05	print_char	6	174	0x00C21E
Variable	0x06	print_num	32	170	0x00C201
Variable	0x07	random	561	156	0x0056D0
Variable	0x08	push	2,970	208	0x00BB08
Variable	0x09	and	1	169	0x00C1FB
Variable	0x09	pull	554	210	0x00BB11
Variable	0x0D	store	233	4,769	0x00B33D
Variable	0x14	add	3	157	0x0056D4
Variable	0x17	div	24	159,448	0x005A92

Replay completion: AwaitingInput at the post-ending prompt. Output confirms the Cryo-Anteroom ending, score 80/80, and no deaths. Route SHA-256: 437621B2651098DB00DFE96723EF348B2DC0785F2114BD548CA8C9B85B88A7B1